

NATIONAL INSTITUTE OF TECHNOLOGY ROURKELA

■ Database Engineering CS2008 ■ End Sem Spring 2020-21 ■ UG ■ Pg 1 ■ FM 50 ■ 2 Hrs

■ Answer ALL questions. ■ All parts of a question MUST be answered together. ■ Mere answers without proper explanation will not fetch marks. ■ Variation in quality of answer will vary the secured marks.

1. ~~a.~~ Does BCNF guaranty zero redundancy?
~~b.~~ What are the problems in a non-normalized database?
c. For a schema R (A, B, C, D) bearing FDs $F=\{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$, find if the decomposition of R into $R_1(A, B, C)$ and $R_2(B, C, D)$ is dependency preserving?
d. For a schema R (A, B, C, D, E, F) bearing FDs $G=\{A \rightarrow B, A \rightarrow C, C \rightarrow D, AE \rightarrow F\}$, find if the decomposition of R into $R_1(A, B, C)$, $R_2(C, D)$, and $R_3(A, E, F)$ is lossless? [1.5+2+4.5+4.5]

2. ~~a.~~ Check if $BD \rightarrow E$, $AC \rightarrow E$ are the redundant FDs in $F = \{A \rightarrow B, C \rightarrow D, BD \rightarrow E, AC \rightarrow E\}$
~~b.~~ Eliminate redundant FDs from $F=\{X \rightarrow Y, Y \rightarrow X, Y \rightarrow Z, Z \rightarrow Y, X \rightarrow Z, Z \rightarrow X\}$ using the Membership algorithm.
~~c.~~ What do you mean by soundness and completeness of Armstrong's axioms? [3.5 + 6 + 3]

3. ~~a.~~ When a language is called *relationally complete*?
b. Find the content of tables G, H, I, J, K, L, M, N from the following tables R and S:

$G \leftarrow R \bowtie_{A=D} S$ $H \leftarrow R \ltimes_{A=D} S$ $I \leftarrow R \bowtie_{A=D} S$ $J \leftarrow R \lhd_{A=D} S$
 $K \leftarrow R \triangleright_{A=D} S$ $L \leftarrow R \bowtie_{A=D} S$ $M \leftarrow R \bowtie_{A=D} S$ $N \leftarrow R \bowtie_{A=D} S$

[3 + 1.5 × 8]

R(A, B, C)			S(D, E, F)		
A	B	C	D	E	F
a ₁	b ₁	c ₁	a ₁	e ₁	f ₁
a ₁	b ₂	c ₁	a ₁	e ₂	f ₃
a ₂	b ₁	c ₁	a ₂	e ₂	f ₂
a ₃	b ₂	c ₂	a ₂	e ₄	f ₄
a ₄	b ₃	c ₃	a ₅	e ₂	f ₃

4. ~~a.~~ Explain the need of *Isolation* and *Durability* property of transaction.
~~b.~~ What are the problems caused by concurrent transactions? Explain them.
~~c.~~ What is lost update problem? Explain with an example.
d. Check if the following schedule S on two transactions T_1 and T_2 is conflict serializable:
 $S : \{ R_1(A), W_1(A), R_2(A), W_2(A), R_1(B), W_1(B), R_2(B), W_2(B) \}$

[(1+1)+1.5+1.5+5]

■ BEST OF LUCK ■